

Integrable Systems — Exercise Sheet 14

The quantum inverse scattering method

Variant: Questions only

Conventions, as in Lecture 14: $\hbar = 1$ except in Exercises 3 (iii), 5 and 6, where $\hbar > 0$. The auxiliary spaces a, b and the sites $n = 1, \dots, N$ are copies of $\mathbb{C}^2$, and P_{xy} exchanges the factors x and y . $R(u) = u - iP$; the Lax operator of the Heisenberg chain is $L_{an}(\lambda) = \lambda + iP_{an}$; the monodromy is $T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a1}(\lambda)$ and the transfer matrix $t(\lambda) = \text{tr}_a T_a(\lambda)$. The RTT relation is $R_{ab}(\lambda - \mu)T_b(\mu)T_a(\lambda) = T_a(\lambda)T_b(\mu)R_{ab}(\lambda - \mu)$, and $C = P_{ab}P_{an} = P_{an}P_{bn} = P_{bn}P_{ab}$, $C^{-1} = P_{an}P_{ab} = P_{ab}P_{bn} = P_{bn}P_{an}$. Classically, $\{f, g\} = \sum_n (\partial_{q_n} f \partial_{p_n} g - \partial_{p_n} f \partial_{q_n} g)$, and legs are as in Lecture 9: $T_1 = T \otimes \mathbb{1}$, $T_2 = \mathbb{1} \otimes T$, $(A \otimes B)_{(ac),(bd)} = A_{ab}B_{cd}$.

Exercise 1 (the Lax operator of the Heisenberg chain). (i) On $\mathbb{C}_a^2 \otimes \mathbb{C}_n^2$ show that $P_{an} = \frac{1}{2}(\mathbb{1} + \sigma_a \cdot \sigma_n)$. (ii) Deduce Lecture 14's matrix form of $L_{an}(\lambda)$ in the auxiliary space, with entries $\lambda + \frac{i}{2} \pm iS_n^\pm$ on the diagonal and $iS_n^\pm$ off it. (iii) Write $L_{an}(\lambda)$ as a 4×4 matrix in the basis $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ (auxiliary state first). Check that exactly six entries are not identically zero (as functions of λ), and explain, with the vertex picture of Lecture 14, why two of them are $\lambda + i$ and not λ . (iv) For which λ is $L_{an}(\lambda)$ not invertible? Show that there it is a multiple of a projector, and that $L_{an}(0) = iP_{an}$.

Exercise 2 (RLL fixes R). Let $R_{ab} = \alpha + \beta P_{ab}$ with numbers α, β (which may depend on λ, μ). (i) Following the proof of Lecture 14's RLL theorem, show that

$$R_{ab}L_{bn}(\mu)L_{an}(\lambda) - L_{an}(\lambda)L_{bn}(\mu)R_{ab} = [\alpha - i\beta(\lambda - \mu)](C - C^{-1}).$$

(ii) Deduce that the non-zero R_{ab} of this form satisfying RLL are exactly the non-zero multiples of $R_{ab}(\lambda - \mu) = \lambda - \mu - iP_{ab}$. (iii) At $\lambda = \mu$ this R is a multiple of P_{ab} . Show directly that $P_{ab}L_{bn}(\lambda)L_{an}(\lambda) = L_{an}(\lambda)L_{bn}(\lambda)P_{ab}$.

Exercise 3 (the order of RTT). Let T satisfy RTT with $R(u) = u - iP$. (i) Using $R(u)R(-u) = -(u^2 + 1)\mathbb{1}$ (Lecture 14), show that for $\lambda - \mu \neq \pm i$ the RTT relation is equivalent to $\tilde{R}_{ab}(\lambda - \mu)T_a(\lambda)T_b(\mu) = T_b(\mu)T_a(\lambda)\tilde{R}_{ab}(\lambda - \mu)$, with $\tilde{R}(u) = u + iP$: the other order of the monodromies, with the other sign in R . (ii) Show that $T'(\lambda) = T(-\lambda)$ satisfies the other order with the same R : $R_{ab}(\lambda - \mu)T'_a(\lambda)T'_b(\mu) = T'_b(\mu)T'_a(\lambda)R_{ab}(\lambda - \mu)$ for all λ, μ (where T is a polynomial in λ , as for the chain). For the Heisenberg chain, what is the Lax operator of T' ? (iii) Classically, let T satisfy Lecture 9's $\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda)T_2(\mu)]$ with $r(\lambda) = -P/\lambda$. Show that $T'(\lambda) = T(-\lambda)$ satisfies the same bracket with $-r$ in place of r . Then, restoring $\hbar$ ($R = \lambda - i\hbar P = \lambda(\mathbb{1} + i\hbar r)$), expand the relation of (ii) to order $\hbar$ as in Lecture 14's classical limit, taking the symbol of $T'_a T'_b - T'_b T'_a$ to be $i\hbar \{T'_1, T'_2\} + O(\hbar^2)$, and check that it gives the same.

Exercise 4 (three sites by hand). Take the Heisenberg chain with $N = 3$. (i) Show that $\text{tr}_a P_{an} = \mathbb{1}$, $\text{tr}_a(P_{an}P_{am}) = P_{nm}$ for $n \neq m$, and $\text{tr}_a(P_{a3}P_{a2}P_{a1}) = P_{32}P_{31}$, and deduce

$$t(\lambda) = 2\lambda^3 + 3i\lambda^2 - \lambda(P_{12} + P_{13} + P_{23}) - iP_{32}P_{31}.$$

(ii) Show that $U = P_{32}P_{31}$ sends $x_1 \otimes x_2 \otimes x_3$ to $x_3 \otimes x_1 \otimes x_2$, so that $t(0) = -iU$ is $-i$ times the shift by one site. Show that U commutes with $P_{12} + P_{13} + P_{23}$, and deduce $[t(\lambda), t(\mu)] = 0$

directly. (iii) Show that $i t(0)^{-1} t'(0) = P_{12} + P_{23} + P_{31}$, and so that Lecture 10's Hamiltonian at $N = 3$ is $H = \frac{i}{2} (i t(0)^{-1} t'(0) - 3)$. (iv) Find the eigenvalues of $t(\lambda)$ and of H , with their multiplicities.

Exercise 5 (symbols). One degree of freedom, $[q, p] = i\hbar$, $p = -i\hbar \partial_q$ on functions of q . (i) Show that $p g(q) = g p - i\hbar g'$, and by induction $p^k g = \sum_{j=0}^k \binom{k}{j} (-i\hbar)^j g^{(j)} p^{k-j}$: the symbol formula of Lecture 14 for $F = p^k$, $G = g$. (ii) Compute $[p^2, e^q]$ exactly, and compare its symbol with $i\hbar \{p^2, e^q\}$. (iii) Show that $[p, g(q)] = i\hbar \{p, g\}$ exactly, for every g . Deduce that the commutators of the entries of the quantum Toda site matrix $\begin{pmatrix} \lambda - p & e^{-q} \\ -e^q & 0 \end{pmatrix}$ are exactly $i\hbar$ times the Poisson brackets of the corresponding functions. Why does (ii) not contradict this?

Exercise 6 (the quantum Toda chain, $N = 2$ and 3). Quantise Lecture 9's periodic Toda chain as in Lecture 14: $[q_m, p_n] = i\hbar \delta_{mn}$, $L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}$, $t(\lambda) = \text{tr}(L_N(\lambda) \cdots L_1(\lambda)) = \sum_k c_k \lambda^{N-k}$. (i) For $N = 2$, find c_1 and c_2 , and show that $H = \frac{1}{2} c_1^2 - c_2 = \frac{1}{2} (p_1^2 + p_2^2) + e^{q_1 - q_2} + e^{q_2 - q_1}$. (ii) Show directly that $[c_1, c_2] = 0$ for $N = 2$. (iii) For $N = 3$, show that $c_3 = -p_1 p_2 p_3 + p_1 e^{q_2 - q_3} + p_2 e^{q_3 - q_1} + p_3 e^{q_1 - q_2}$, and explain why no ordering question arises. Then verify $[H, c_3] = 0$ by hand, with $H = \frac{1}{2} \sum_n p_n^2 + e^{q_1 - q_2} + e^{q_2 - q_3} + e^{q_3 - q_1}$, and check that the terms of order $\hbar^2$ cancel too.

Exercise 7 ★. Any spin. Let $\mathbf{S}$ be spin operators on a space V , $[S^j, S^k] = i\epsilon_{jkl} S^l$, and on $\mathbb{C}_\alpha^2 \otimes V$ put $L_{\alpha n}(\lambda) = \lambda + \frac{i}{2} + i \sigma_\alpha \cdot \mathbf{S}$. Lecture 14 cites that it satisfies RLL with the same R ; prove it. (i) With $X = \sigma_a \cdot \mathbf{S}$ and $Y = \sigma_b \cdot \mathbf{S}$, show that RLL for all λ, μ is equivalent to $[X, Y] = (Y - X)P_{ab}$. (ii) Prove this identity, using only the commutation relations of $\mathbf{S}$. (iii) For an irreducible spin $s \geq \frac{1}{2}$, find the eigenvalues of $L_{\alpha n}(\lambda)$ and their multiplicities, and show that for $s = \frac{1}{2}$ you recover $\lambda + iP_{\alpha n}$.